# CampusNotes - Academic Study Resource

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING - SEMESTER 5

**OFFICIAL ACADEMIC STUDY MATERIAL - CSE CURRICULUM**

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## Assignment 2: Normalization  3NF, BCNF, 4NF  & Decomposition

Course: Advanced Database Engineering [23CS3503]
Category: Assignments | Assignment #2 | Academic Year: 2026-27

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### 1. Overview & Objective:

Database normalization theory, dependency preservation, and lossless decomposition.

### 2. Assignment Problems & Instructions:

1. Determine candidate keys from a set of functional dependencies using Armstrong axioms.
2. Compute the canonical cover  minimal cover  of a given functional dependency set.
3. Decompose relation R into BCNF verifying lossless join and dependency preservation.
4. Explain Multi-Valued Dependencies  MVD  and normalize relations into 4NF.

**Submission Guidelines & Verification:**

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